

Indian Institute of Technology Kharagpur
Department of Mathematics
MA11003 - Advanced Calculus
Problem Sheet - 9
Autumn 2025

1. Solve the following differential equations:

- (a) $(x^2 D^2 - 3xD + 4)y = 2x^2$
- (b) $(x^2 D^2 + 7xD + 13)y = \log x$
- (c) $(x^2 D^2 - 4xD + 6)y = x^4$
- (d) $(x^2 D^2 + xD - 1)y = x^m$, where $m \neq \pm 1$
- (e) $(x^2 D^2 - 3xD + 5)y = \sin(\log x)$
- (f) $(x^2 D^2 - xD + 1)y = 2 \log x$
- (g) $(x^2 D^2 - (2m - 1)xD + (m^2 + n^2))y = n^2 x^m \log x$
- (h) $(x^2 D^2 - xD + 2)y = x \log x$
- (i) $(x^2 D^2 - 3xD + 5)y = x^2 \sin(\log x)$
- (j) $(x^4 D^4 + 6x^3 D^3 + 9x^2 D^2 + 3xD + 1)y = (1 + \log x)^2$

2. Solve the following differential equations:

- (a) $(1 + x)^2 y'' - 4(1 + x)y' + 6y = 6(1 + x)$
- (b) $(x + 1)^2 y'' + (x + 1)y' = (2x + 3)(2x + 4)$,
- (c) $(1 + 2x)^2 y'' - 6(1 + 2x)y' + 16y = 8(1 + 2x)^2$

3. Apply the method of variation of parameters to solve the following differential equations:

- (a) $y'' - 2y' = e^x \sin x$
- (b) $y'' - 6y' + 9y = \frac{e^{3x}}{x^2}$
- (c) $y'' - 2y' + y = e^x \log x$
- (d) $y''' + y' = \tan x$
- (e) $y''' - 2y'' - 21y' - 18y = 3 + 4e^{-t}$
- (f) $y'' - 2y' + 2y = e^x \tan x$

4. Using the method of variation of parameters, solve

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} = e^x \sin x$$

with $y(0) = 0$ and $(\frac{dy}{dx})_{x=0} = 0$